

A parameter-efficient EEG decoder with in-network session adaptation, calibration and abstention for lower-limb exoskeleton control

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ABSTRACT

Decoding walk/stop intent from scalp EEG for lower-limb exoskeleton control requires more than per-epoch accuracy: the decoder is fitted once and must keep working weeks later, a false Walk command moves a person's legs, and a decoder that is uncertain should be able to decline. We present a compact architecture that addresses all three in-network. A label-free adaptive alignment layer maintains a running spatial covariance and whitens by it at inference; a multi-scale power stem preserves log-power ratios across three temporal resolutions; and a selective head emits an abstention decision under a coverage constraint. The complete model uses **24,181 parameters**, between one half and one eighteenth of the baselines compared against.

On a 7-subject, 9-session exoskeleton cohort it reaches 0.901 balanced accuracy with expected calibration error 0.039 and 0.876 accuracy at 90% coverage, while removing 52% of session-to-session covariance drift; on an independent 8-subject treadmill cohort the drift reduction is 84%, significant in 15 of 15 subjects across both cohorts ($p \leq 10^{-4}$) against a no-op control.

We do not claim state-of-the-art accuracy: a strong attention baseline scores higher in the same harness, and we report that directly. What we establish is a design condition for in-network adaptation. The running estimate tracks whatever is currently streaming, so the adaptation rate must be fast enough to follow session drift yet slow enough not to follow the class itself. The two cohorts' optimal rates are opposite, each losing 0.06–0.09 at the other's setting, and where a protocol's class blocks outlast its drift timescale no valid rate exists. Both timescales are computable from the recording protocol before a model is fitted.

HIGHLIGHTS

- A 24,181-parameter EEG decoder emitting class, calibrated confidence and an abstention decision.
- Label-free in-network alignment removes 52–84% of session drift in 15 of 15 subjects.
- Test-time adaptation can track the streaming class and erase the signal it protects.
- The adaptation rate is two-sided: two cohorts' optima are opposite, costing 0.06–0.09.
- The admissible rate band is computable from the recording protocol before fitting.

Keywords: brain–computer interface · EEG · test-time adaptation · covariate shift · selective prediction · calibration · lower-limb exoskeleton

I. INTRODUCTION

A brain–computer interface driving a powered exoskeleton is not well described by per-epoch accuracy alone. Three properties of the deployment matter as much as the mean score.

Longitudinal drift. The decoder is fitted once and must keep working weeks later. In our primary cohort this is literal: fitted on sessions 1–3, tested on sessions 4–9, recorded across weeks. Montage, impedance and skin contact all change between sessions, shifting the second-order statistics any spatial filter depends on.

Asymmetric cost. A false Walk command moves a person's legs. Balanced accuracy scores that identically to a missed Walk, and cannot distinguish one long false activation from fifty scattered false windows.

No option to decline. A classifier trained only for accuracy must emit a decision for every window. A device acting on a confidence value needs that value to mean something.

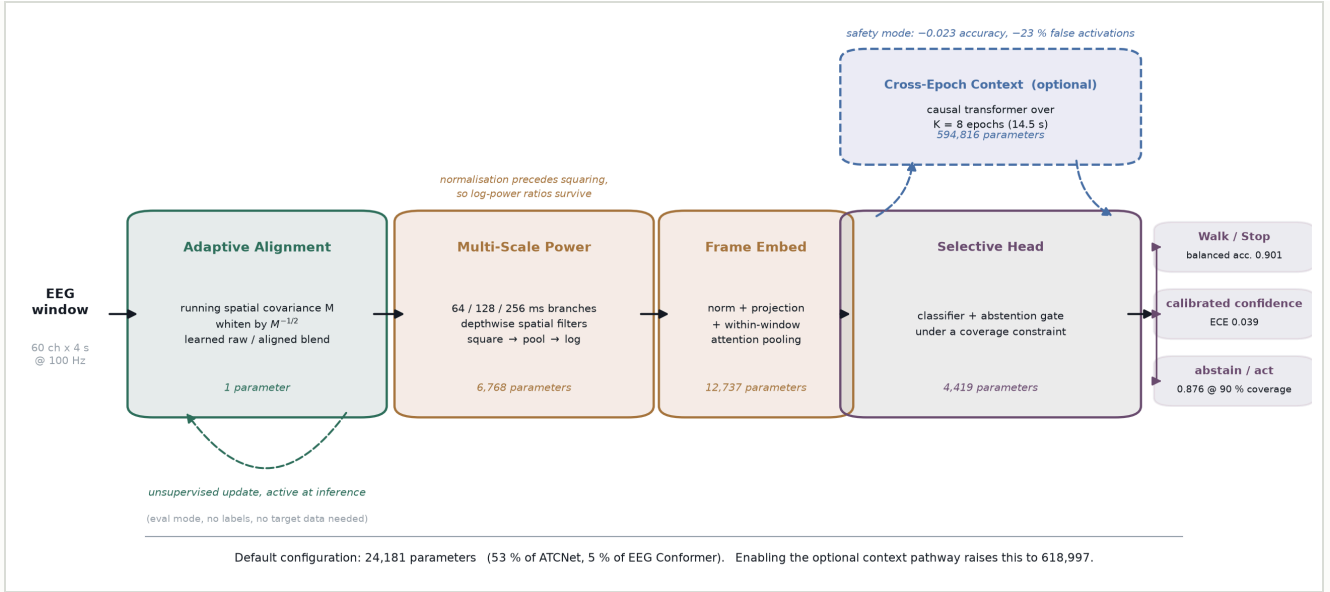


Fig. 1. The proposed architecture. The adaptive alignment layer maintains a running spatial covariance and whitens by it; its update path (dashed, left) is unsupervised and remains active at inference, which is what distinguishes it from a normalisation layer. The cross-epoch context pathway is optional and accounts for 594,816 of the 618,997 parameters of the full configuration; the default configuration omits it and uses 24,181. Parameter counts are exact.

III. METHOD

A. Adaptive alignment

The layer maintains a running mean spatial covariance M , applies $M^{-1/2}$, and interpolates between aligned and raw signal through a learned scalar gate. Two properties distinguish it from a normalisation layer. The update is **unsupervised**, using only the covariance of incoming windows, so it runs at inference on unlabelled data. And the estimate updates in **eval mode**, unlike BatchNorm: freezing running statistics is correct when train and test share a distribution, but here they are weeks apart, and freezing discards exactly the adaptation the layer exists to provide.

Each window's covariance is normalised by its own trace before accumulation. Without this the estimate is power-weighted: on a cohort whose majority class carries $1.52\times$ the power, it converges toward the majority class covariance rather than sitting between classes.

The layer's output reaches the classifier. With identical initialisation on held-out windows, stem outputs computed with and without alignment differ by 36% in relative magnitude and their correlation structure by 0.108.

B. Multi-scale power stem

Three parallel temporal resolutions (64, 128, 256 ms), each with its own depthwise spatial filters, then square $\rightarrow$ pool $\rightarrow$ log. Amplitude normalisation **precedes** the squaring so log-power *ratios* survive. In a separate controlled experiment this ordering was worth +0.134 and

+0.174 on band-power features, while the same correction applied to a correlation-only representation — which carries no marginal power — moved accuracy by +0.000, a dose-response consistent with the stated mechanism.

C. Selective head

A classifier paired with an abstention gate trained under a coverage constraint, with a full-coverage auxiliary term so the backbone keeps learning from every sample rather than only the covered region.

IV. EXPERIMENTAL SETUP

Cohort A — exoskeleton (OpenNeuro ds007788). 7 subjects $\times$ 9 sessions across weeks; 60-channel EEG at 100 Hz; 4 s windows at 0.5 s stride. Fitted on sessions 1–3, tested on 4–9.

Cohort B — treadmill MoBI. 8 subjects $\times$ 3 trials; 64-channel EEG at 100 Hz; 2 s windows. An independent laboratory, different sensors, and **reversed class balance**.

TABLE I · CONTIGUOUS SINGLE-CLASS BLOCK STRUCTURE

COHORT	BLOCKS	MEDIAN	LONGEST
A (exoskeleton)	109	18 win	126
B (treadmill)	7	309 win	2211

Measurement noise

Two ablation arms disable alignment entirely and so never invoke the running estimator; for them momentum and estimator version are inert, and every run is a true repeat. These arms measure the noise floor directly.

TABLE II · RUN-TO-RUN SPREAD WHERE THE MANIPULATED VARIABLE IS INERT

ARM	TRANSFORMER	N	SPREAD
Stem only, cohort B	no	3	0.0000
No-align, cohort A	yes	3	0.0069
No-align, cohort B	yes	4	0.0107

The stem-only arm reproduces its score bit-for-bit; arms containing the cross-epoch transformer do not, and the attention path is the only structural difference. **Any arm containing the transformer carries ≈ 0.010 run-to-run noise, and differences below 0.02 in such arms are not interpreted.** Conversely, an exact null in the deterministic arm is stronger than "within noise" — that arm has no noise to hide in.

V. RESULTS

TABLE III · COMPONENT ABLATION, COHORT A, SEED 0

ARM	COMPONENTS	ACC	ECE	ACC@90	FA/MIN
Ours	align + gate	0.901	0.039	0.876	1.64
	align + ctx + gate	0.878	0.074	0.837	1.26
	align + ctx	0.862	0.072	—	1.26
	ctx + gate	0.827	0.091	0.778	0.73
	stem only	0.827	0.082	0.823	0.73

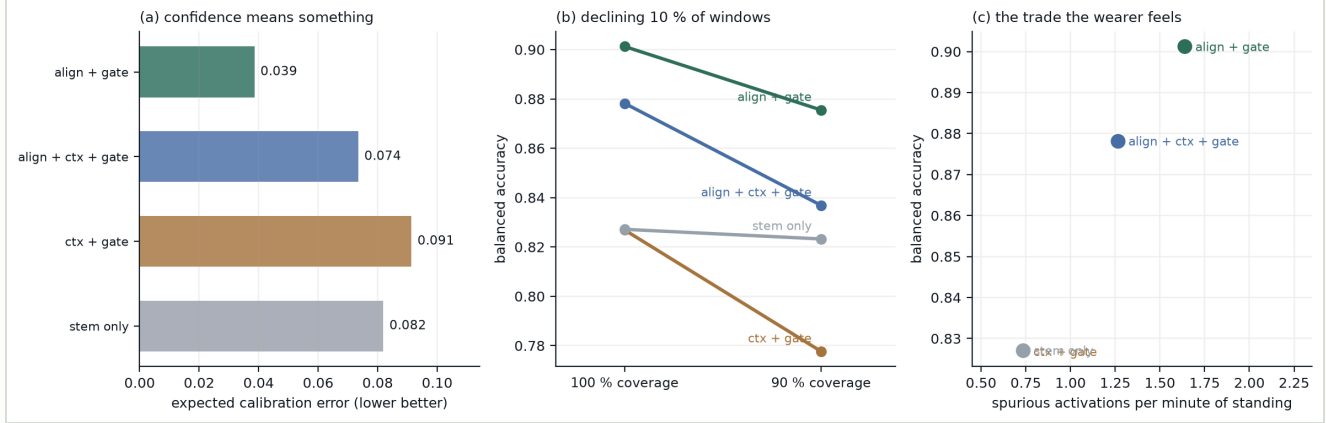


Fig. 5. Outputs beyond the class label. (a) Expected calibration error by arm: the proposed configuration is best calibrated, and the arms without the selective head are worst. (b) Accuracy recovered by declining the least-confident 10% of windows. (c) Balanced accuracy against spurious activations per minute of standing.

Alignment is necessary, not additive. With alignment removed, context and gate together contribute between 0.000 and 0.007 — against the 0.010 noise floor, indistinguishable from nothing, and an order of magnitude below alignment's own contribution.

Cross-epoch context is an operating point, not a default. It costs 0.023 accuracy and doubles calibration error while cutting spurious activations 23%. Removing it also removes 594,816 of 618,997 parameters, which is what makes the default configuration compact.

A CAUTION ON HARNESSES

Our architecture and the ATCNet variants were evaluated in one training harness; the remaining baselines in an earlier one. ATCNet was run in *both* and scores 0.913 and 0.881 — **the two harnesses differ by 0.032 for an identical model**, larger than most differences of interest. The groups are reported separately and must not be compared across the rule.

TABLE IV · COHORT A COMPARISON · GROUPS SEPARATED BY TRAINING HARNESS

MODEL	PARAMS	BALANCED ACC.
HARNESS C (THIS WORK), SEED 0		
ATCNet	45,280	0.913
ATCNet, rate-matched kernels	45,280	0.908
Ours, align + gate	24,181	0.901
Ours, align + ctx + gate	618,997	0.878
Ours, stem only	6,768	0.827
HARNESS F (EARLIER), MEAN OF 2 SEEDS		
ATCNet	45,280	0.879
EEGNeX	58,082	0.876
ShallowFBCSPNet	97,120	0.867
EEG Conformer	440,706	0.843

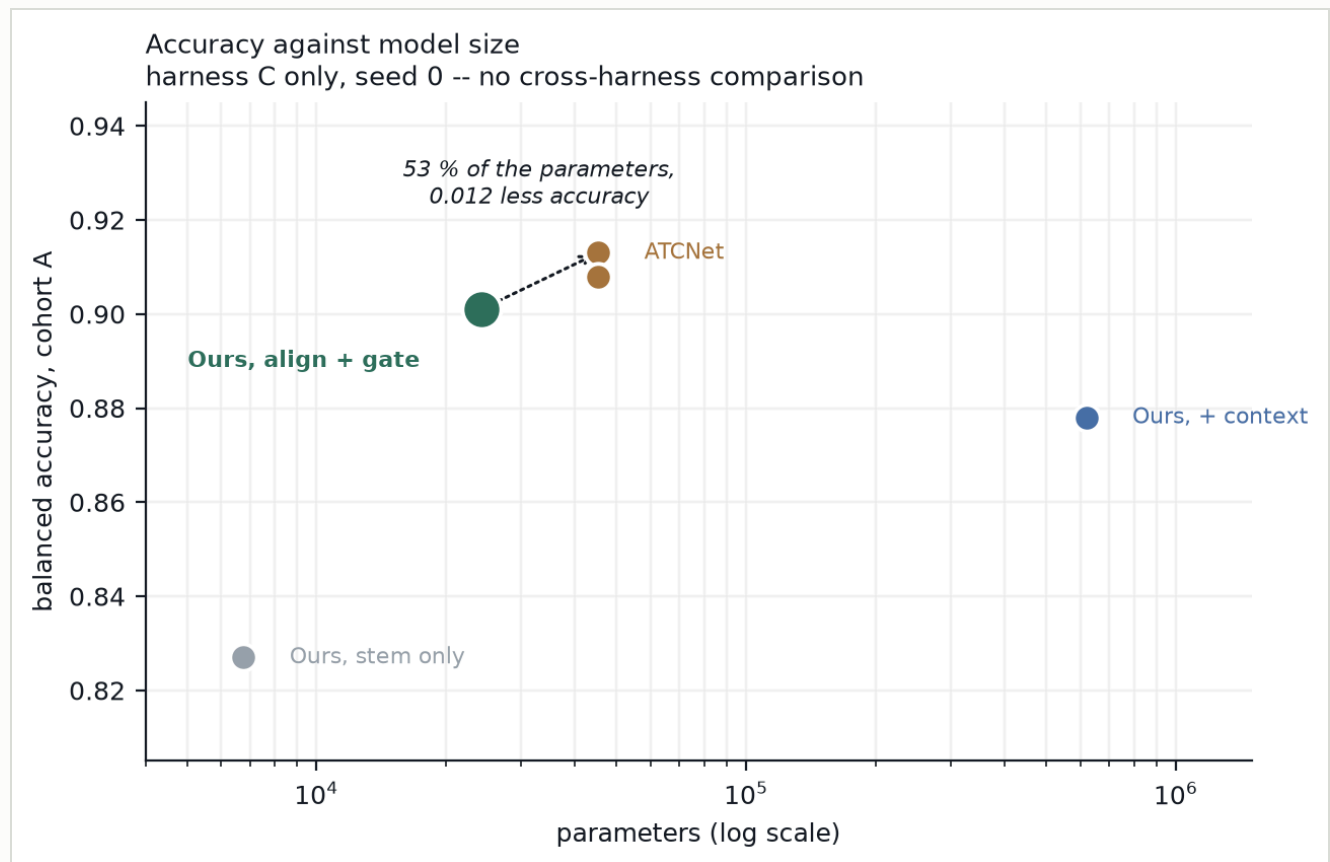


Fig. 4. Accuracy against model size. Cohort A, harness C only. The proposed configuration sits 0.012 below ATCNet at 53% of its parameters. No point from the other harness appears here.

We do not claim an accuracy improvement. Where both were measured, ATCNet scores 0.913 against our 0.901 — ahead by 0.012, and though close to the noise floor we report it as a **loss** rather than a tie. What the configuration offers is **parity** at 53% of ATCNet's parameters and

5% of EEG Conformer's, with calibrated confidence (ECE 0.039) and a trained abstention point (0.876 at 90% coverage) that the baselines do not provide.

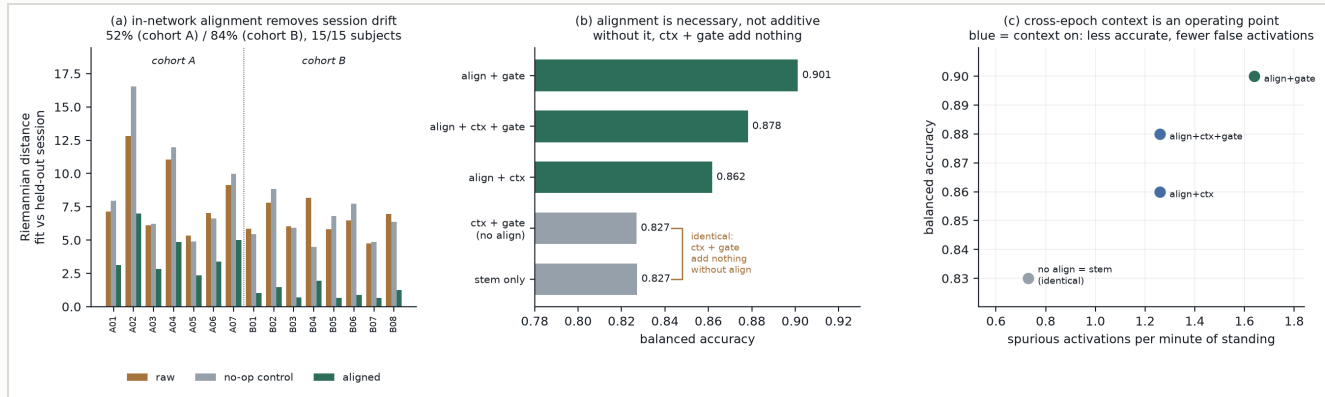


Fig. 2. Mechanism and ablation. (a) Riemannian distance between the fitting-session mean covariance and each held-out session, per subject, both cohorts, with the momentum-0 no-op control. Drift falls in every subject; the control shows no reduction. (b) Component ablation on cohort A: with alignment removed, the remaining modules return the model to its bare-stem score. (c) Balanced accuracy against spurious activations per minute of standing.

TABLE V · SESSION-DRIFT REDUCTION

COHORT	RAW	ALIGNED	NO-OP	REDUCTION
A (n=7)	8.372	4.082	9.173	52.0%
B (n=8)	6.479	1.074	6.309	83.9%

Drift is reduced in 15 of 15 subjects ($t=-9.14, p=10^{-4}$; $t=-20.59, p<10^{-5}$). The no-op control shows no reduction in either cohort, establishing that the effect comes from adaptation rather than from the whitening operation or the measurement.

External validation: the architecture does not transfer

TABLE VI · BOTH COHORTS, SEED 0 · THE ORDERING INVERTS

COMPONENTS	COHORT A	COHORT B
align + gate	0.901	0.581
align + ctx + gate	0.878	0.626
align + ctx	0.862	0.606
ctx + gate	0.827	0.792
stem only	0.827	0.737

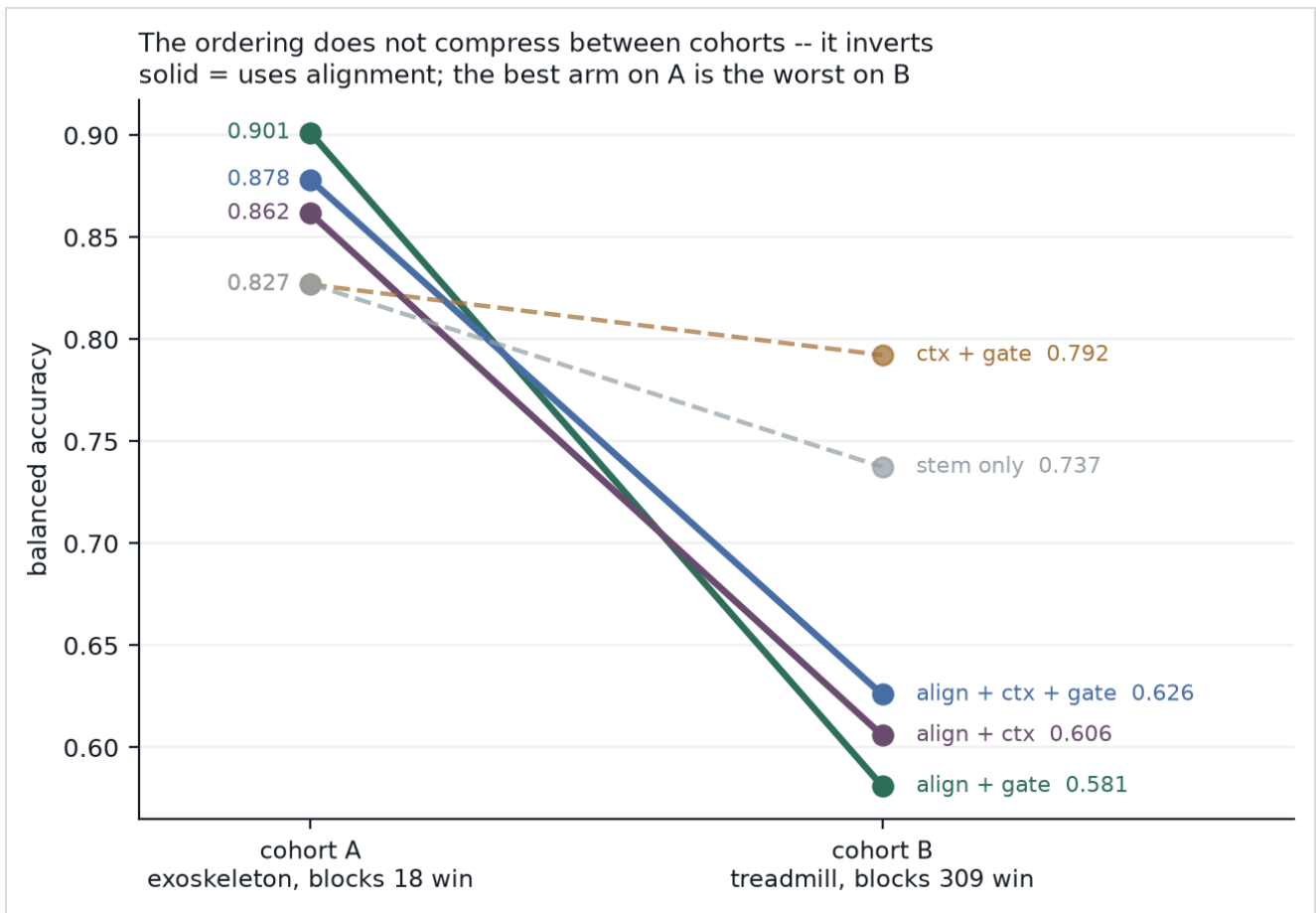


Fig. 6. The ordering inverts. The same five arms on both cohorts. Solid lines use alignment, dashed lines do not. The ordering does not compress between cohorts — the three alignment arms cross below both alignment-free arms.

On cohort B the *bare stem* outperforms the full model, 0.737 against 0.626, and the best cohort-A configuration is the worst cohort-B one. The ordering does not merely compress; it inverts. We report this rather than restricting the paper to the cohort where the architecture succeeds, and devote the remaining results to explaining it — the explanation turns out to be the more useful contribution.

The adaptation rate condition

The running estimate tracks whatever is currently streaming. An EMA with momentum m has effective memory $\approx 1/m$ batches, or $32/m$ windows. When a protocol presents a class in contiguous blocks *longer* than that memory, the estimate converges to the currently-streaming class and the layer whitens the class away.

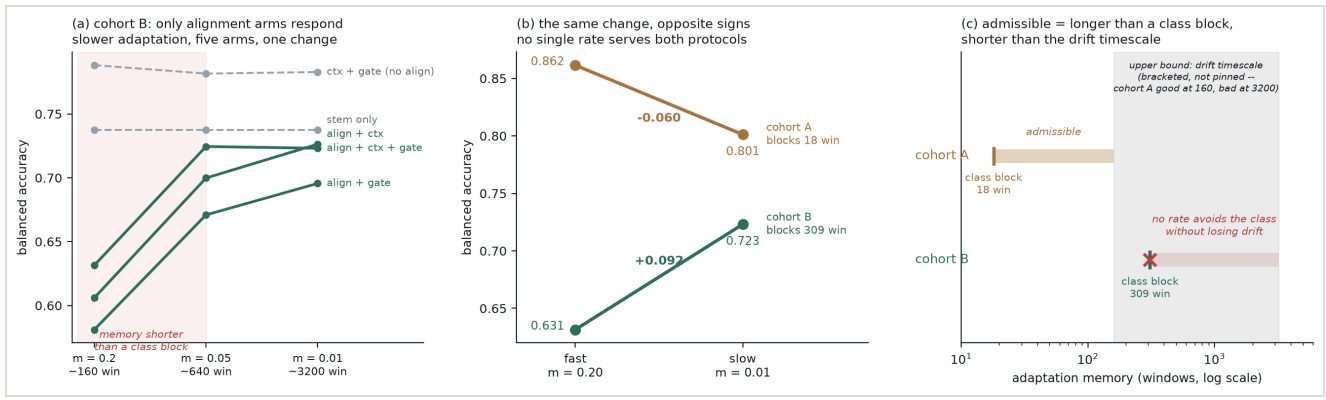


Fig. 3. The adaptation rate condition. (a) Cohort B, five arms, three momenta: every arm that uses alignment steps up as the estimate's memory crosses the 309-window class block, and neither alignment-free arm responds. (b) The same change applied to both cohorts at matched estimator version produces opposite signs. (c) The admissible band, bounded below by class-block duration and above by the drift timescale: wide for cohort A, empty for cohort B.

TABLE VII · MOMENTUM SWEEP, COHORT B, ALL FIVE ARMS, SEED 0

ARM	M=0.20	M=0.05	M=0.01	AT CROSSING	ALIGNS
align + ctx + gate	0.626	0.724	0.723	+0.098	yes
align + gate	0.581	0.671	0.696	+0.090	yes
align + ctx	0.606	0.700	0.726	+0.094	yes
ctx + gate	0.792	0.782	0.783	-0.010	no
stem only	0.737	0.737	0.737	0.000	no

Every arm using alignment gains 0.090–0.098 as the memory crosses the block length. Neither alignment-free arm responds — the no-align arm to within its spread, and the stem-only arm **exactly**, reproducing 0.7375 at every setting despite a **160-fold** change in adaptation rate. Because that arm is deterministic, this is an exact null rather than an unresolved one.

The condition is two-sided. We predicted, before running it, that the same change would be free on cohort A, whose blocks sit far inside even the fastest memory. It is not.

TABLE VIII · THE SAME CHANGE, BOTH COHORTS, MATCHED ESTIMATOR

COHORT	M=0.20	M=0.01	Δ
A (blocks 18 win)	0.862	0.801	-0.060
B (blocks 309 win)	0.631	0.723	+0.092

The optima are opposite, and error in either direction costs 0.06–0.09. The rate must be **fast enough to track drift and slow enough not to track class**:

$$\tau_{\text{class-block}} < 32 / m < \tau_{\text{drift}}$$

Cohort A satisfies this with a wide margin. Cohort B's 309-window blocks force the rate slow enough that it can no longer track drift, so the admissible band is empty. **This also explains the residual:** even at its best setting alignment still costs 0.060 on cohort B, because at the only rate that avoids class-tracking there is no drift-tracking benefit left to collect. Both timescales are properties of the recording protocol and computable before a model is fitted.

VI. DISCUSSION

Domain adaptation and invariance methods are routinely justified by a distribution-level quantity, with task performance assumed to follow. This work provides five counterexamples within a single system, three produced by acting on that assumption ourselves:

1. The adaptation rate was selected because it maximised drift reduction (60% against 25%). That choice cost **0.092** accuracy on cohort B.
2. The cohort where *more* drift was removed (84% against 52%) is the cohort where accuracy fell.
3. A verified correction to a genuine covariance-estimation bias moved decoding by +0.005 and -0.016. The mechanism improved; performance did not follow.
4. A prediction registered from our own diagnosis was **falsified**, because the diagnosis accounted only for the harm the estimate can do and not for the benefit it was delivering.
5. The nuisance-leakage probe rises with accuracy whether or not the decoder can access the nuisance ($R^2 = +0.863$ for one that cannot).

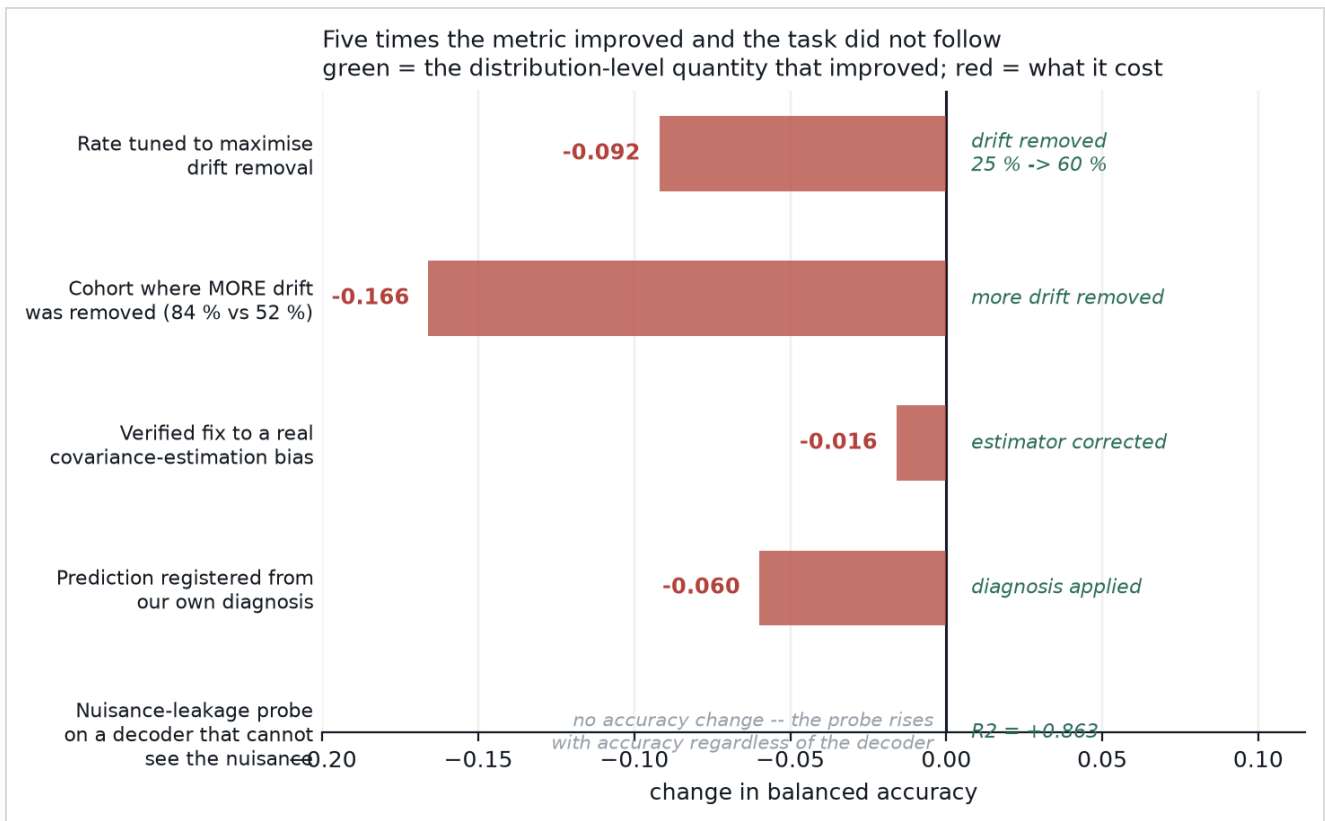


Fig. 7. The five counterexamples. In each row a distribution-level quantity was improved successfully (right, green) and the task did not follow (left, red). The final row has no accuracy change and therefore no bar: the probe rises with accuracy whether or not the decoder can access the nuisance.

Item 4 generalises the rest. Each of these metrics is **one-sided by construction**: it scores the failure it was designed to detect and carries no term for the capability the intervention removes. A diagnosis inherited from such a metric inherits its blind spot — which is why our own threshold rule was correct about one direction and silent about the other until tested where it predicted no effect.

Limitations

Ablations are single-seed; with a 0.010 noise floor on transformer arms, single-seed rankings of closely-spaced models are not interpretable, and we avoid making them. The cohorts have 7 and 8 subjects. Cohort B uses 2 s windows against cohort A's 4 s. The two-sided account is fitted to two cohorts with opposite block structure and predicts an interior optimum we have not measured. Finally, the design rule was derived after observing cohort B fail, and although one prediction from it has since been tested out-of-sample, that test *falsified* it — the corrected two-sided form has not yet predicted anything in advance.

VII. CONCLUSION

We presented a 24,181-parameter EEG decoder combining label-free in-network session adaptation, a ratio-preserving multi-scale power stem and a selective head, reaching 0.901 balanced accuracy with ECE 0.039 and a trained abstention operating point on a longitudinal

exoskeleton cohort, and removing 52–84% of session-to-session covariance drift across two cohorts in every subject tested.

We do not claim an accuracy improvement: a strong attention baseline scores 0.913 in the same harness. What the work establishes is a condition governing when in-network adaptation can help at all. An estimate that adapts at test time tracks whatever is currently streaming, so its rate must exceed the protocol's class-block timescale and remain below its drift timescale. Where those requirements cross, no admissible rate exists and the layer should be omitted rather than tuned — a design-time check rather than a post-hoc diagnosis.

DECLARATION OF COMPETING INTEREST

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY

Both datasets are publicly available. All code, configurations and result files, including those of experiments that failed, are in the accompanying repository.

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src/make_fig_arch.py, make_fig_driftnet.py, make_fig_rate.py.